

Offensive Systems and Play-Calling Languages

Why two teams can run the same pass play under completely different names, what each naming system is built to do, and how to tell whose language an offense speaks from the couch.

Two broadcasts on the same Sunday. In one, a microphone in the huddle catches the quarterback repeating a call that runs past twenty words: a formation, a motion, a protection, and then a separate instruction for almost every receiver. In the other, the quarterback barely huddles; he walks to the line, says one word to the receivers on his right and another to the receivers on his left, and the ball is snapped. Then something odd happens on the replay: the two offenses run *the same pass play*. Same routes, same depths, same read. Why would two professional staffs give one play such different names, and does it matter? It matters a great deal. The way a team names its plays decides how fast it can play, how much it can teach a rookie in a week, and how easily it can change course in the third quarter. This chapter is about those languages, the offensive families that grew up around them, and how to tell which one a team speaks by watching it.

You'll be able to...

- Translate one pass play into the three classic NFL languages: West Coast words, Air Coryell route numbers and Erhardt-Perkins concept names.
- Say what each major family (West Coast, Air Coryell, Erhardt-Perkins, run-and-shoot, Air Raid, Shanahan/McVay, spread option) is built to do, and what it gives up.
- Explain why the naming system shapes installation, tempo and in-game adjustment.
- Describe how a system is installed: base plays first, then tags, then the weekly game-plan packages that become the call sheet.
- Fingerprint an offense from the couch (under center or shotgun, motion, splits, empty, huddle) and guess its family.
- Use the “checklist so far” card for Parts 1–4.

You'll need

This chapter closes the passing part of the book, so it leans on most of it. One line each:

- **Play calls** (What a Play Really Is): a play call is a sentence (formation, motion, protection, concept, tags); a team's words are its *verbiage*; a *tag* changes one player's job; to *install* is to teach a set of plays. That chapter also introduced the three big NFL naming families in one paragraph; here we build them properly.
- **Routes and reads** (Passing Fundamentals): the *route tree* is the standard menu of routes; an *option route* (or *choice route*) lets the receiver pick his break after the snap; a *progression* is the quarterback's fixed order of looks.
- **Concepts** (Dropback Concepts): a *pass concept* is a set of routes designed as a unit, usually to stretch one or two defenders vertically or horizontally. Mesh and Y-cross, the Air Raid's signature plays, are drawn there.
- **Quarterback play** (Quarterback Play): how the quarterback works through his reads, moves in the pocket and manages risk; and the “system quarterback” debate, which this chapter's systems sit underneath.
- **Formations and motion** (Personnel Groupings, Spread Formations, Motion and Shifts): personnel numbers (“11” is one back and one tight end), 2x2 and 3x1 (trips), *empty* (no back), *condensed* splits (receivers tight to the line), and why offenses move players before the snap. Receivers are counted from the sideline in: #1 is the widest on his side, then #2, then #3.

What a system is (and what it isn't)

Broadcasters say “system” constantly (“he's a system quarterback,” “they run a West Coast system”) and rarely say what they mean. Here is a working definition. An **offensive system** is a coherent way of playing offense, and it has four layers:

1. **A belief about what to attack.** Walsh's 49ers believed short, accurate, on-time throws could do the work of a running game. Don Coryell's Chargers believed in stretching the field vertically. The Shanahan school believes everything should start from the run.
2. **A core of plays that express the belief.** Usually a few dozen pass concepts and a handful of run schemes, drilled until they are automatic.

3. **A language** for naming formations, motions, protections and plays: the verbiage from Part 1.
4. **A method of teaching it:** what gets installed first, how practice is organized, what the quarterback is allowed to change at the line.

What a system is *not*: a formation. Every NFL system lines up in shotgun trips and under-center two-tight-end sets. It is not a list of plays either: two playbooks can contain nearly identical plays and still be different systems, because the words, the teaching and the priorities differ. And it is not one coach's private property. Systems spread through coaching trees (assistants who carry a head coach's methods to new jobs; the trees themselves are mapped in the Walsh case study and Appendix E), and every carrier changes it a little.

This chapter concentrates on the third layer, the language, for a practical reason: the language is the part that shapes what you see on TV. A long call takes time to say, and that time shows up on the play clock. A one-word call can be shouted at the line, so the offense can skip the huddle. A language that ties routes to positions makes it awkward to move players around, so the formations stay simpler. Where these systems *came from*, and what football problem each was invented to solve, is the subject of 1978 and the Passing Revolution and The College Laboratory; here we name them and learn to recognize them.

! Why it exists

Why does an offense need a *system* at all, rather than just a big collection of good plays? Because eleven people have to execute every play in about three seconds, with forty seconds to hear it, and a roster turns over every year. A system compresses. If every pass concept follows the same rules (the same depths, the same rules for adjusting a route against man or zone, the same protection words), a player who learns ten words knows a hundred plays, and a new receiver signed in October can play on Sunday. The price is coherence's usual price: a system is good at what it was built for and awkward at everything else, and defenses study it too.

i Madden vs. real life

A video-game playbook is organized by formation: pick "Gun Trips TE," then pick one of fifteen plays drawn underneath it. The plays have names, but the names are labels, not language: nothing about "PA Boot Over" tells a player what to do if you move him to the other side of the formation. A real playbook is organized more like a grammar. The formation words, motion words, protection words and concept words are taught separately and combined at the line, which is why a real offense can run a play it has never run before (a familiar concept from a new formation) without anyone learning anything new. The menu in the game is the *output* of a system; the system is the thing that generates it.

Three ways to say one play

The fastest way to see what a play-calling language does is to say one play in several languages. Here is a simple three-receiver pattern from shotgun trips: the outside receiver runs deep, the middle one runs an out at about eight yards, the inside one (the tight end) runs to the flat, and the receiver alone on the left side runs a dig. The protection is the same six-man "60" in every version, so only the route words differ.

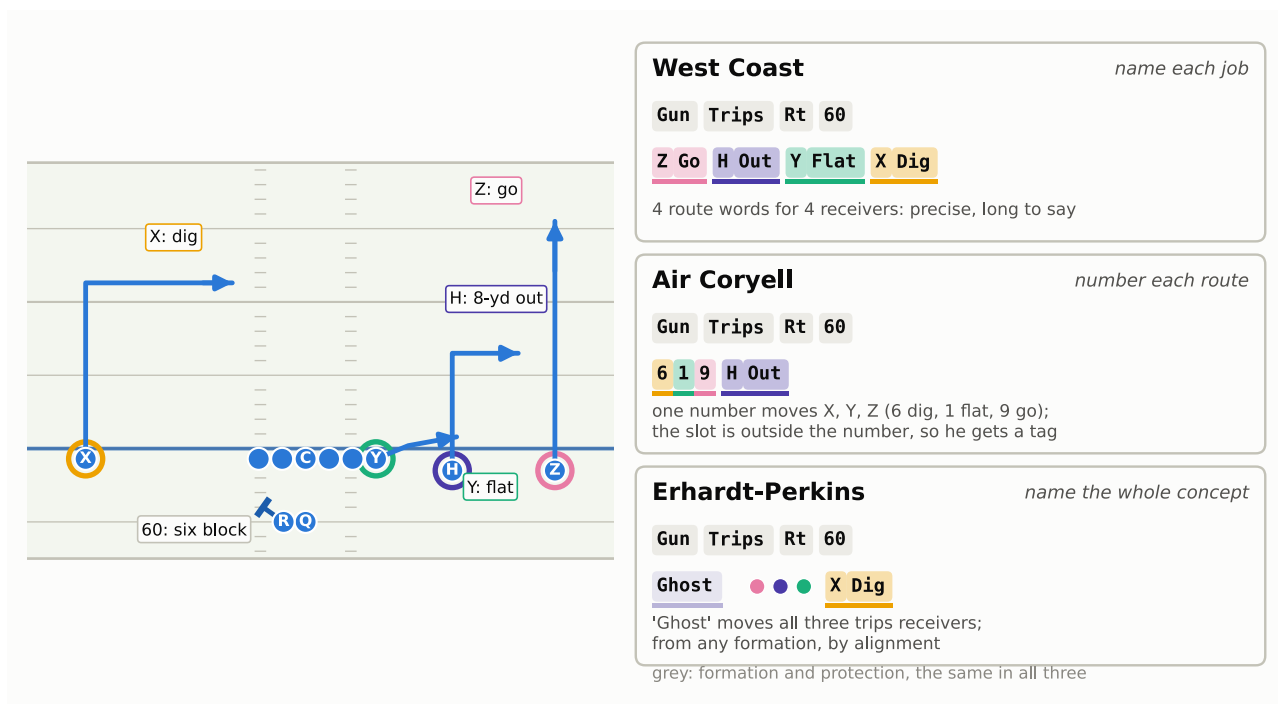


Figure 1: One play, three languages (the calls are illustrative, written in each family's style). Left: the play. Right: the call each family would use, with each word colored by the receiver it moves. The West Coast call names every receiver's route. The Coryell call packs three receivers into one three-digit number, read in a fixed order (X, then Y, then Z), and tags the fourth. The Erhardt-Perkins call gives the three-receiver side one concept word, 'Ghost', and lets alignment decide who runs what.

Read the three calls against the picture.

West Coast: name every job. The call lists the formation and protection, then a word for each receiver: Z Go, H Out, Y Flat, X Dig. Nothing is left to inference. If the coach wants the out at seven yards instead of eight, or the flat route to start with a fake block, there is a word for that too, and it goes in the call. The cost is length: four route words here, and real West Coast calls carry far more. Chris Brown's example of how unwieldy they can get is a real call, "Scatter-Two Bunch-Right-Zip-Fire 2 Jet Texas Right-F Flat X-Q": two formation and motion words, a protection, a play name and two individual assignments.¹ Many West Coast plays are named for one featured route ("Texas" is the back's angle route), and the rest of the players simply memorize what they do on that play.

Air Coryell: number every route. The digits come from a numbered route tree, and their order says who runs them: here the first digit is X's route, the second Y's, the third Z's. "619" is X on a 6 (the dig), Y on a 1 (the flat), Z on a 9 (the go). One short number moves three receivers. The slot isn't in the three-digit slot, so he gets a word of his own.

Erhardt-Perkins: name the concept. "Ghost" is a word every receiver knows: the outermost receiver on that side runs vertical, the next one breaks to the outside at about eight yards, the innermost runs straight to the flat.² It doesn't say *who*. Whoever lines up outermost runs the vertical, so the same word works from trips, from a bunch, or with a running back as the third man. That is **concept-based terminology**: a word for a whole pattern, with each player's role decided by where he lines up. The lone receiver on the other side still needs instructions, which here come as a tag. Real Erhardt-Perkins calls often pair two concepts, one for each side ("Ghost/Tosser," with Tosser a pair of slants on the other side).

None of these is "better." Each is a different bet about where the hard part of offense is: in the precision of each route (West Coast), in speed and simplicity of the call (Coryell), or in flexibility across formations and personnel (Erhardt-Perkins). Most modern playbooks mix all three, as Part 1's example "Stick X Dig" did. But each family still has a native tongue, and its accent shows on the field.

¹Chris B. Brown, "Speak My Language," *Grantland* (Jan. 25, 2013), on West Coast calls in which "passing plays are denoted by the primary receiver's route while other players memorize their tasks," with the example call quoted here: <https://grantland.com/features/how-terminology-erhardt-perkins-system-helped-maintain-dominance-tom-brady-patriots/>

²Brown, "Speak My Language" (*Grantland*, 2013), defining Ghost (outermost receiver vertical, an inside receiver breaking outside at about eight yards, the innermost to the flat), Tosser (double slants) and the paired "Ghost/Tosser" call: <https://grantland.com/features/how-terminology-erhardt-perkins-system-helped-maintain-dominance-tom-brady-patriots/>

The West Coast offense: say exactly what you mean

The **West Coast offense (WCO)** is the family descended from **Bill Walsh**, who built its foundations as an assistant with the Cincinnati Bengals in the early 1970s and made it famous as head coach of the San Francisco 49ers (1979–88), winning three Super Bowls.³ Walsh was a former high-school coach and lifelong teacher who thought of football as a craft to be broken into precise, rehearsable parts, and the WCO is his temperament in scheme form. In Cincinnati, with a quarterback who threw accurately but not far, he leaned on short, quick, horizontal throws timed to the quarterback's footwork, and used them the way other teams used runs: as a reliable way to gain five yards on first down. (How that happened, and why it is called "West Coast" at all, a name Walsh himself didn't love, is told in 1978 and the Passing Revolution; the full offense, play by play, is the Walsh 49ers case study.)

The WCO language follows from that belief. If the whole system rests on *timing* (the receiver breaks at exactly this depth, the quarterback's back foot hits at exactly that moment, the ball is out before the break is finished), then precision is everything, and a call should leave nothing to interpretation. So West Coast calls:

- **Name the formation and backfield in detail**, often with colors or words for where the backs stand, and a separate word for any motion.
- **Name a protection**, often with a number and a word ("2 Jet").
- **Name a play**, frequently for the featured route of one player, with the rest of the routes memorized as part of that play.
- **Add individual assignments** for anyone doing something different from the base version ("F Flat," "X Q").

This is why the WCO is famous for long calls, and why its descendants still have them. A Kyle Shanahan call quoted by a 49ers beat reporter runs about twenty words.⁴ Jon Gruden, a Holmgren and Walsh-tree coach, became a minor celebrity on ESPN partly for reciting WCO calls with relish. The verbiage is a badge of the family, and the coaches who speak it tend to defend it on the same ground Walsh did: the long call is long because it is exact.

What it buys:

- **Precision and detail.** The coach can adjust any one player's job in the call itself, without a separate meeting.
- **A huge, finely graded menu.** Because the parts are named individually, the same concept can carry many small variations, each with its own name.
- **Teaching through the call.** Players learn the system by hearing it; the call is the assignment.

What it costs:

- **Time.** A long call takes several seconds to say in the headset and again in the huddle, which makes the no-huddle awkward. Shanahan, as Part 1 noted, prefers spoken calls to wristbands for exactly this reason: he wants the full call heard and repeated.
- **Memory.** A rookie, or a veteran from another system, faces hundreds of words, and memorized "the rest of you know what to do" plays. The WCO has a reputation as one of the hardest systems to learn quickly.
- **Brittleness when you move people.** Words like "X Dig" tie an assignment to a person. Put X in the slot and some of the logic has to be re-taught.

Common misconception

"West Coast offense" does not mean "short passes." Walsh's offense threw deep when the defense allowed it, and its descendants (Mike Holmgren's Packers, Andy Reid's Eagles and Chiefs, the Shanahan family) have thrown plenty of long balls. What the WCO did was treat the short, horizontal, on-time throw as a *base* play rather than a desperation play, and build a language and a teaching method precise enough to make those throws reliable. The label now describes a family resemblance (timing, horizontal stretches, long precise calls) more than a fixed scheme.

Air Coryell: routes as numbers

Air Coryell is named for **Don Coryell**, head coach of the San Diego Chargers from 1978 to 1986, whose Dan Fouts-led offenses led the league in passing yards year after year.⁵ Coryell came out of San Diego State, where he had won with a pass-first college

³Pro Football Hall of Fame, "Bill Walsh" (head coach 1979–88, three Super Bowl titles): <https://www.profootballhof.com/players/bill-walsh/> ; Bill Walsh with Brian Billick and James Peterson, *Finding the Winning Edge* (Sports Publishing, 1998), on his years as a Bengals assistant under Paul Brown (1968–75) and the short, timing-based passing game he developed there.

⁴NBC Sports Bay Area, "Kyle Shanahan's long play-call shows why 49ers' playbook is 'no joke'" (June 16, 2020): <https://www.nbcsportsbayarea.com/nfl/kyle-shanahans-long-play-call-shows-why-49ers-playbook-is-no-joke/1264096>

⁵Pro Football Hall of Fame, "Don Coryell" (Chargers head coach 1978–86; inducted 2023): <https://www.profootballhof.com/players/don-coryell/> .

team, and he took Sid Gillman's idea of attacking the whole field vertically and made it louder. Where Walsh thought of the pass as a horizontal substitute for the run, Coryell thought of it as a vertical weapon: receivers running hard down the field, throws timed to land in the gaps between deep defenders.

The language that grew around it is the **digit system**: every route in the tree has a number, and a pass play is a string of digits, one per receiver, in a fixed order.

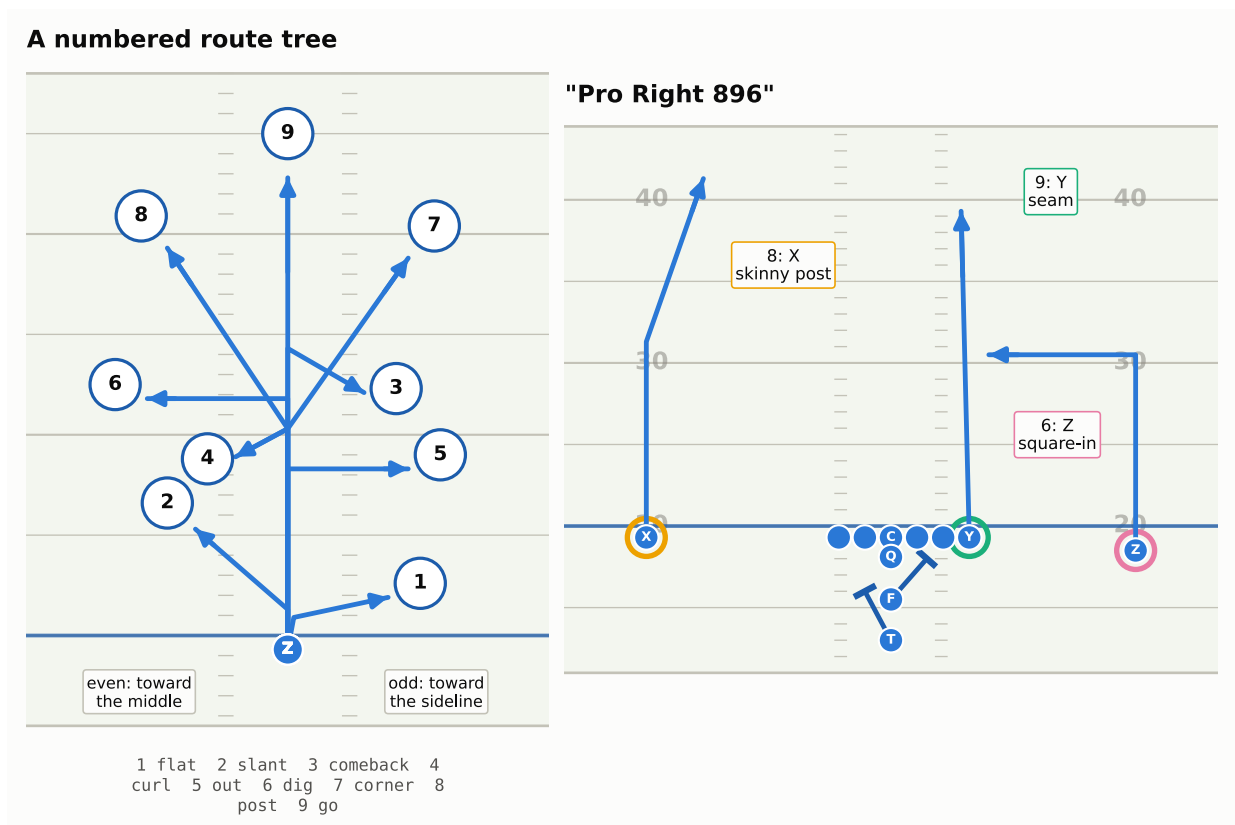


Figure 2: The digit system. Left: one common numbered route tree, drawn for a receiver on the right side (so 'outside' is to the right). Odd numbers break toward the sideline, even numbers toward the middle, and the numbers grow with depth, ending at 9, the go. Right: the call '896' from a pro set: X (left) runs an 8, a skinny post; Y, the tight end, runs a 9 up the seam; Z runs a 6, a square-in underneath Y. Trees and digit orders vary by team; this is one widely taught version.

Chris Brown's well-known example is exactly this call: in "896," the split end (X) runs a skinny post, the tight end (Y) runs straight up the seam, and the flanker (Z) runs a square-in underneath.⁶ Three digits, three receivers, and a play that attacks the deep middle at two levels. Running backs and slot receivers, who don't fit in the three-digit order, get tags or default routes. The system's descendants (Ernie Zampese, Norv Turner's Cowboys of the early 1990s, Mike Martz's "Greatest Show on Turf" Rams of 1999–2001) kept the numbers and the appetite for vertical routes.⁷

What it buys:

- **Speed and compactness.** "896" is three syllables that move three people. It can be said in the headset in under a second and signaled with fingers.
- **Infinite combinations.** Nine routes for three receivers is 729 combinations before any tags, and every one of them is instantly understandable to anyone who knows the tree.
- **A vertical bias that suits the tree.** Because the tree's numbers rise with depth, the natural calls are combinations of deep routes, which is what Coryell wanted.

What it costs:

⁶Brown, "Speak My Language" (Grantland, 2013): in "896," the "8" sends the split end on a skinny post, the "9" sends the tight end straight up the seam, and the "6" sends the flanker on a square-in. Route trees and the digit order differ from team to team: <https://grantland.com/features/how-terminology-erhardt-perkins-system-helped-maintain-dominance-tom-brady-patriots/>

⁷Tim Layden, *Blood, Sweat and Chalk* (Sports Illustrated Books, 2010), on the Coryell lineage through Ernie Zampese, Norv Turner and Mike Martz.

- **The numbers only describe single routes.** A digit says nothing about how two routes work *together*. The concept (this route clears space for that one) lives in the coaches' heads and the meeting room, not in the call.
- **Position-bound.** The order of the digits is tied to X, Y and Z. Move the tight end into the backfield or the flanker into the slot and the numbers have to be re-mapped, or the play needs extra words.
- **A tree is not a playbook.** Good combinations have to be found and taught; a bad combination is just as easy to call as a good one.

Erhardt-Perkins: one word, any formation

Erhardt-Perkins is named for **Ron Erhardt** and **Ray Perkins**, who developed it on Chuck Fairbanks's New England Patriots staff in the 1970s.⁸ Its line of descent runs through the 1980s New York Giants, where Perkins and then Bill Parcells were head coaches with Erhardt as offensive coordinator, to Parcells's assistants, among them Bill Belichick and Charlie Weis, and from Belichick's Patriots to **Josh McDaniels**, who called New England's offense for most of the Tom Brady years and again in 2025.⁹

The idea is the one we saw in "Ghost": name the **concept**, not the routes, and let each receiver's alignment tell him his role. The concept vocabulary is the system. Ghost is a vertical, an eight-yard out and a flat; Tosser is a pair of slants; and the full Erhardt-Perkins vocabulary holds dozens more, often paired one per side of the field. Charlie Weis put the point simply: "You can cut down on the plays and get different looks from your formations and who's in them. It's easier for the players to learn."¹⁰

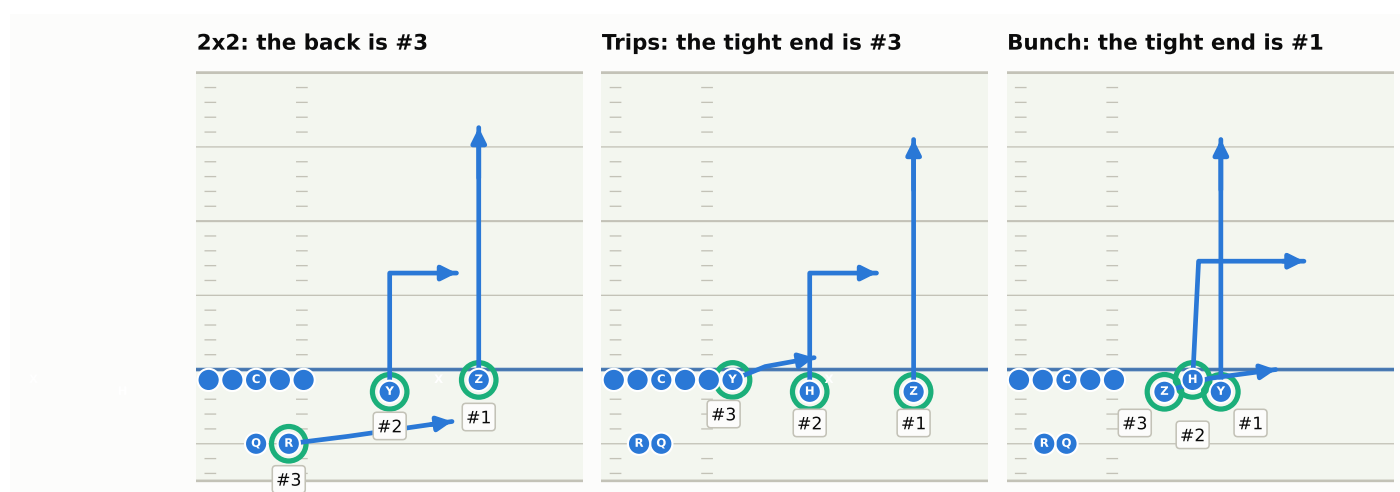


Figure 3: Why concept words travel. The same call, 'Ghost', from three formations. The outermost receiver on the right (#1) runs vertical, the next (#2) runs the eight-yard out, the innermost (#3) runs to the flat. Who those players are changes completely: in the 2x2 set the running back is #3; in trips it's the tight end; in the bunch the tight end is outermost and runs the vertical. Nobody learned a new play.

What it buys:

- **Portability.** A concept learned once can be run from any formation and with any personnel. That lets the offense show the defense many different looks while asking its own players to learn very little. It also suits a team that changes personnel often: a tight end, a slot receiver or a running back can play the "#3" role without learning a new play.
- **Short calls.** One or two words per side, which can be said at the line. Brown points out that in 2012 the Patriots led the league in total plays, first downs, points and yards, by wide margins, while playing fast with one- or two-word calls; the terminology is what made that tempo possible.¹¹
- **Game-plan agility.** Each week the staff can dress the same concepts in new formations, so the defense's film study shows them looks they haven't seen even though the players are running plays they've known for years.

⁸Brown, "Speak My Language" (*Grantland*, 2013), on Erhardt and Perkins developing the system on Chuck Fairbanks's Patriots staff in the 1970s, Perkins's later head-coaching jobs (Giants, Alabama, Buccaneers), and the line through Parcells to Belichick's Patriots: <https://grantland.com/features/how-terminology-erhardt-perkins-system-helped-maintain-dominance-tom-brady-patriots/>

⁹New England Patriots coaching biography of Josh McDaniels (offensive coordinator 2006–08 and 2012–21; returned as offensive coordinator in 2025): <https://www.patriots.com/team/coaches/>; 2025 staff as compiled in this course's fact sheet (checked 2026-10-06).

¹⁰Charlie Weis, quoted in Brown, "Speak My Language" (*Grantland*, 2013): <https://grantland.com/features/how-terminology-erhardt-perkins-system-helped-maintain-dominance-tom-brady-patriots/>

¹¹Brown, "Speak My Language" (*Grantland*, 2013): "In 2012, the Patriots were the league leaders in total plays, first downs, points, and yards — all by a significant margin," with the terminology credited for letting them play no-huddle on one- or two-word calls: <https://grantland.com/features/how-terminology-erhardt-perkins-system-helped-maintain-dominance-tom-brady-patriots/>

What it costs:

- **Less control over details.** “Ghost” says *what kind* of play it is, not exactly how deep each route is against every coverage. The details have to live in rules the players have learned, and in tags.
- **A big vocabulary of a different kind.** The words are fewer per call but there are many concepts to learn, and each comes with rules for every role. Players coming from position-based systems say the adjustment is real.
- **Everyone has to know every role.** If alignment decides the job, a receiver must know all three Ghost roles, and the quarterback must know where each will be from every formation.

i Film room: Patriots 2025 — McDaniels’s language with a second-year quarterback

The Erhardt-Perkins line is still live. In 2025, Mike Vrabel’s first season as New England’s head coach, Josh McDaniels returned as offensive coordinator to a team with second-year quarterback Drake Maye. The Patriots went 14–3, reached Super Bowl LX (losing 29–13 to Seattle), Maye finished second in the MVP voting by one first-place vote, and McDaniels was named the league’s Assistant Coach of the Year.¹²

What to notice: the concept language is a *teaching* language as much as a calling one. A young quarterback learning an Erhardt-Perkins offense learns patterns, the way you learn chords before songs, and then sees the same few patterns again and again from different formations. When you watch a McDaniels offense, look for the same two- or three-man combination (a vertical with an out under it, a pair of slants, a stick route with a flat) showing up from a different formation each time. That repetition across disguises is the system’s signature. (On the 2025 Patriots’ tempo and huddle habits, see the fingerprint chart later in this chapter.)

The run-and-shoot: let the receivers decide

The three NFL families so far all assume the coach decides each receiver’s route before the snap, even if some routes have built-in options. The **run-and-shoot** goes further: it gives the decision to the receivers. Its bones came from **Glenn “Tiger” Ellison**, an Ohio high-school coach who, facing a losing season in 1958, rebuilt his offense around the way kids play in the sandlot (spread out, run around, get open) and published it in 1965 as *Run and Shoot Football: Offense of the Future*. **Darrel “Mouse” Davis**, a small, loud, endlessly inventive coach, read the book and spent four decades turning it into a modern passing system at Portland State, in the USFL and in the NFL.¹³

¹²Course fact sheet (checked 2026-10-06): New England 14–3 in 2025; Super Bowl LX, Seahawks 29, Patriots 13 (Feb. 8, 2026); Matthew Stafford edged Drake Maye for MVP by one first-place vote; Josh McDaniels named AP Assistant Coach of the Year. NFL.com, “List of NFL Honors award winners from 2025 NFL season”: <https://amp.nfl.com/news/list-of-nfl-honors-award-winners-from-2025-nfl-season> ; https://en.wikipedia.org/wiki/Super_Bowl_LX

¹³Glenn “Tiger” Ellison, *Run and Shoot Football: Offense of the Future* (Parker Publishing, 1965); *Sports Illustrated*, “19th Hole: The readers take over — Ellison’s offense” (Aug. 18, 1986): <https://vault.si.com/vault/1986/08/18/19th-hole-the-readers-take-over-ellisons-offense> ; American Football International, “After decades, the inventors of the run and shoot share all of their secrets”: <https://www.americanfootballinternational.com/?p=120709> ; on Ellison’s 1958 Middletown (Ohio) season and the “Lonesome Polecat” formation: https://en.wikipedia.org/wiki/Spread_offense

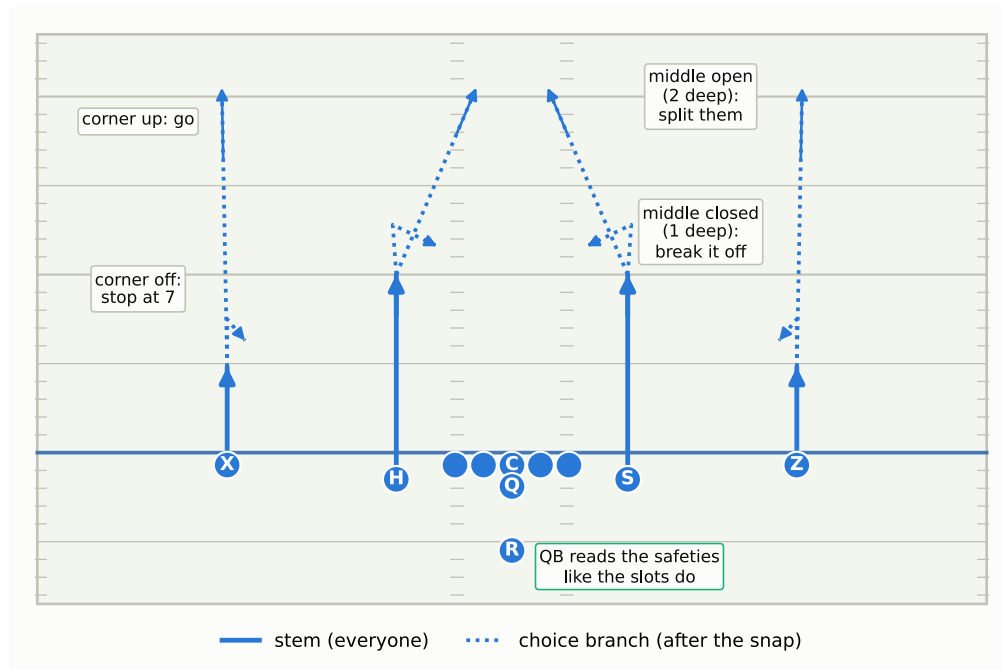


Figure 4: The run-and-shoot's base look and its choice routes (no defense shown, because the routes depend on it). Four receivers, one back, no tight end: two split ends (X, Z) on the line and two slotbacks (H, S) just off it outside the tackles. Solid lines are the stem every receiver runs; dotted lines are the branches he chooses between after the snap. The slots read the middle of the field: with two deep safeties and the middle open, split them; with one safety in the middle, break it off in the open window in front of him. The outside receivers read their corner. The quarterback reads the same defenders, so he and his receivers reach the same answer.

The run-and-shoot has no tight end and usually one back. Two receivers split wide and two “slotbacks” line up just off the line outside the tackles, which puts four receivers in position to run routes immediately. Pre-snap motion (one slot jogging across the formation) is a staple, both to make the defense show its coverage and to give the motion man a running start. Then comes the core idea: Chris Brown summarizes it as “instead of a single assignment, each wide receiver was given a decision tree.”¹⁴ A slot running a “Go” route might run deep, might stop and turn back to the quarterback, or might run 10 or 12 yards and break across the field, depending on what the defense does. The quarterback reads the same things, often while rolling the pocket toward one side, and throws to the answer.

The language is correspondingly small. If the receivers are doing the deciding, the call only needs to name the family of routes (“Go,” “Choice,” “Switch”), the motion and the protection. Every route has branches, so a handful of plays covers every coverage.

What it buys: a very small playbook, every pass play effective against every coverage, and four receivers who are always threats. It spread the field before “spread” was a common word.

What it costs: pass protection without a tight end, a quarterback who must read on the move, and above all *agreement*. If the receiver reads Cover 3 and breaks it off while the quarterback reads Cover 2 and throws the post, the ball goes to the defense. The system lives or dies on receivers and passers who see the same picture.

¹⁴Chris B. Brown, “Victor Cruz, the New York Giants, and Shades of the Run-and-Shoot Offense,” *Grantland* (Jan. 23, 2012), including the route families “streak,” “switch,” “go” and “choice” and the description of the slot’s options on a “go” route: <https://grantland.com/features/chris-brown-victor-cruz-new-york-giants/>

i Film room: Oilers, Lions and Falcons, early 1990s — the run-and-shoot’s NFL moment

For a few years around 1990 the NFL had a genuine run-and-shoot wave: Warren Moon’s Houston Oilers with their four-receiver attack, the Detroit Lions (with Mouse Davis on the staff before the 1990 season), and the Atlanta Falcons’ “Red Gun” under coordinator June Jones, a Davis disciple who had played quarterback for him at Portland State.¹⁵ The Lions won 12 games in 1991; the Oilers made the playoffs every year they ran it and fielded a top-10 offense each season.¹⁶

What to notice: the pure system faded in the NFL within a few years (critics pointed at sacks, red-zone struggles and the lack of a tight end), but its *ideas* won. The choice route, the receiver who adjusts his break to the coverage, is now standard everywhere; the Patriots’ option routes and the slot receiver who “reads the safety” on a vertical are run-and-shoot grandchildren. Kevin Gilbride, the Oilers’ coordinator in that era, later carried several run-and-shoot favorites into Tom Coughlin’s Giants offenses, which won Super Bowls after the 2007 and 2011 seasons.¹⁷

Here is a choice route against a defense that keeps one safety in the middle of the field.

¹⁵Hawaii Warrior World, “More run-and-shoot: Mouse’s offense clicked,” on Davis at Portland State with quarterbacks June Jones and Neil Lomax: <https://www.hawaiwarriorworld.com/football/more-run-and-shoot-mouses-offense-clicked/>.

¹⁶Brown, “Victor Cruz, the New York Giants, and Shades of the Run-and-Shoot Offense” (*Grantland*, 2012), on the Lions’ 12 wins in 1991, the Falcons’ playoff seasons and the Oilers’ playoff streak and top-10 offenses: <https://grantland.com/features/chris-brown-victor-cruz-new-york-giants/>

¹⁷Brown, “Victor Cruz, the New York Giants, and Shades of the Run-and-Shoot Offense” (*Grantland*, 2012), on Kevin Gilbride’s path from Houston’s coordinator to Tom Coughlin’s Jaguars and Giants and the Giants’ use of “several old run-and-shoot favorites”; Giants’ Super Bowl XLII (2007 season) and XLVI (2011 season) titles: <https://www.profootballhof.com/football-history/super-bowl-history/>

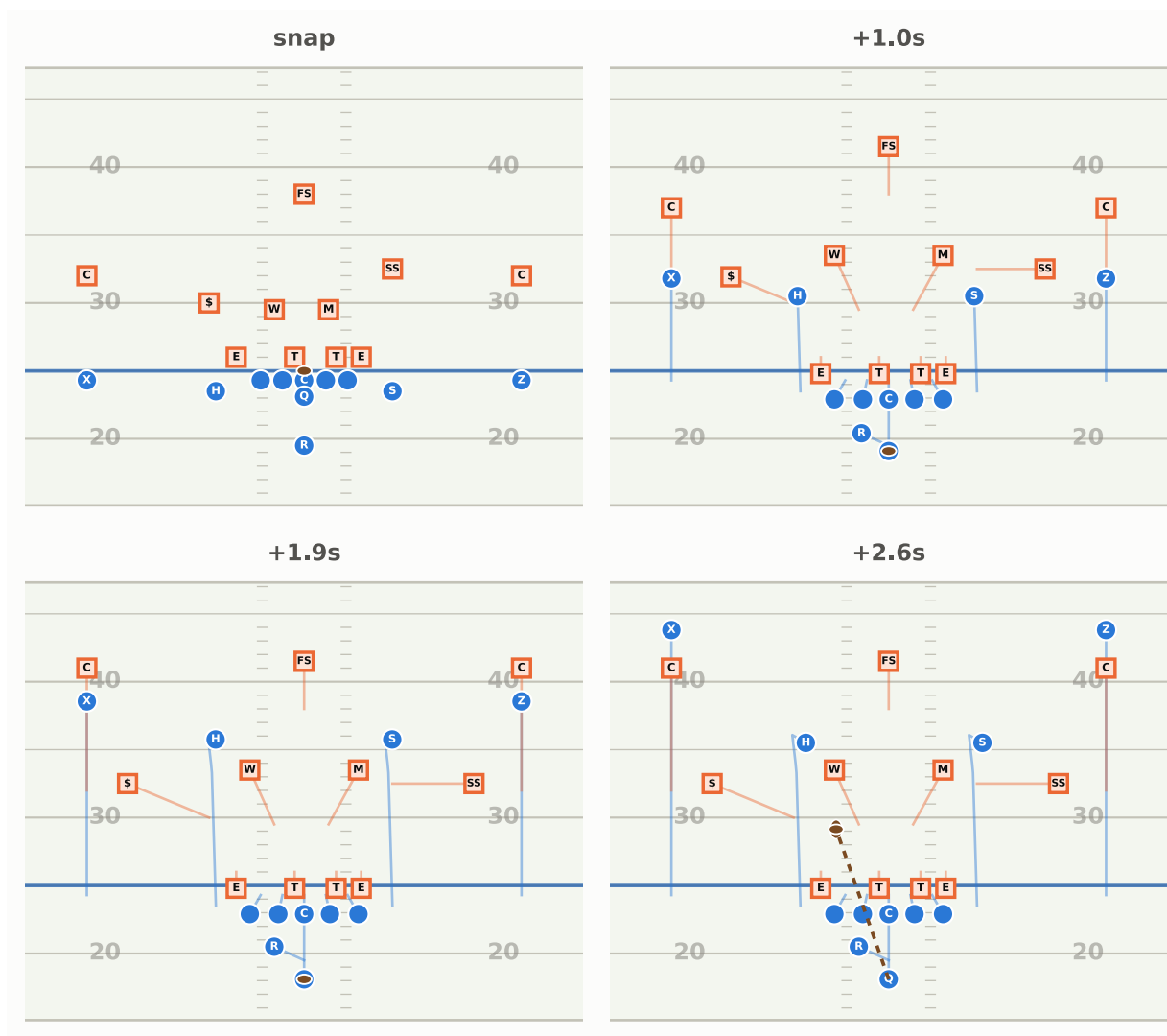


Figure 5: A slot's choice route against one deep safety (simulated, BDB-format). Before the snap the defense shows one safety in the middle. At the snap four defenders rush and seven drop: three deep, four underneath. The left slot (H, ringed) stems vertically, sees the middle safety sitting over the top, and breaks it off at about 12 yards in the window between the hook defender and the curl-to-flat defender. The quarterback, reading the same safety, throws there. Against two deep safeties he would have kept running and split them. In print, the brown dashes are the ball in the air.

The Air Raid: a tiny playbook, repeated forever

The **Air Raid** is the college offspring of all of the above. **Hal Mumme** and his young offensive coordinator **Mike Leach** built it at Iowa Wesleyan, a small college coming off an 0–10 season, starting in 1989, then took it to Valdosta State and the University of Kentucky in the 1990s. Leach, a law-school graduate with a famously wandering mind (pirates, Geronimo, grizzly bears), became its best-known evangelist as head coach at Texas Tech, Washington State and Mississippi State, until his death in December 2022.¹⁸ They borrowed BYU's dropback passing game under LaVell Edwards, pieces of the run-and-shoot and other short passing games, and boiled them down.

The Air Raid's language is defined by subtraction:

- **Few concepts.** A small number of pass plays (mesh, Y-cross, four verticals, a shallow cross, a stick or snag concept, a screen or two) and a couple of runs. Mesh and Y-cross, its signatures, are drawn in Dropback Concepts.
- **Few formations.** Mostly spread 2x2 and 3x1 sets with wide splits.

¹⁸*Sports Illustrated*, "From His Air Raid Offense to His Eccentric Personality, Mike Leach Was One-of-a-Kind" (Dec. 13, 2022): <https://www.si.com/college/2022/12/13/mike-leach-death-legacy-air-raid-college-football> ; Rivals, "The Air Raid Chronicles: Part 1 — The Origins" (Iowa Wesleyan's 0–10 1988 season, Mumme and Leach's arrival in 1989, BYU and run-and-shoot influences): <https://westvirginia.rivals.com/news/the-air-raid-chronicles-part-1-the-origins> ; S. C. Gwynne, *The Perfect Pass* (Scribner, 2016). Leach's jobs: Valdosta State and Kentucky (OC under Mumme), Texas Tech (HC 2000–09), Washington State (2012–19), Mississippi State (2020–22): https://en.wikipedia.org/wiki/Air_raid_offense

- **Short calls.** A word or a number per play, easily signaled from the sideline, so the offense can skip the huddle.
- **Repetition instead of variety.** The same plays are practiced against every coverage, every day, until quarterback and receivers react rather than think. Mumme’s teams famously worked without a thick playbook.

The bet is the opposite of the West Coast’s. The WCO says: build a huge, precise menu and pick the right item. The Air Raid says: run a few things so well that the right item doesn’t matter much. Its NFL footprint is real but indirect. Mesh is in every playbook. Kliff Kingsbury, who played quarterback for Leach at Texas Tech and later coached Patrick Mahomes there, brought Air Raid habits to the Arizona Cardinals as head coach (2019–22) and to Washington as offensive coordinator (2024–25): his offenses lived in the shotgun and led the league in no-huddle snaps, as the chart in *What a Play Really Is* showed.¹⁹

Shanahan and McVay: the run game writes the passing game

The most influential NFL family of the last decade is a dialect of the West Coast offense spoken by **Mike Shanahan**, his son **Kyle**, and the coaches who worked for them: Sean McVay (who coached under Kyle in Washington before taking over the Rams in 2017), Matt LaFleur, Mike McDaniel, Kevin O’Connell (a McVay assistant) and others. Mike Shanahan learned the WCO as the 49ers’ offensive coordinator in the early 1990s and married it, in Denver, to an outside-zone running game; Kyle’s offenses inherited both.²⁰ The case study is The Shanahan Wide-Zone Tree; here we care about the language and what it is for.

The language is long, WCO-style verbal calls, and for a specific reason. The system’s central idea is that **every play should start from the same picture**. The quarterback under center, a fullback or second tight end, receivers in tight splits, someone in motion, and then either an outside zone run, a play-action pass that looks like it, or a bootleg off it (Play-Action, Bootlegs, and Screens). To make a run and three different passes look identical for the first second, every player needs detailed instructions about what to *pretend* as well as what to do: which way to step, how far to sell the run block before releasing, where to be when the quarterback comes out of his fake. That detail goes in the call. The famous Shanahan call that runs about twenty words is long because it describes a play and a disguise at once.

The family’s fingerprints are among the easiest to see on TV: lots of snaps from under center, lots of pre-snap motion (often across the formation at the snap), lots of play-action, receivers lined up close to the formation, and versatile fullbacks and tight ends who block and catch.

The tree also shows how fast “who runs what” changes, which is why every staff in this book is labeled with a season. In 2025, Shanahan (49ers), McVay (Rams), LaFleur (Packers), O’Connell (Vikings) and McDaniel (Dolphins) were all calling plays as head coaches, and Klint Kubiak, son of long-time Shanahan lieutenant Gary Kubiak, called Seattle’s offense on the way to winning Super Bowl LX. By 2026 McDaniel had been fired and become the Chargers’ coordinator, Kubiak was the Raiders’ head coach, and Mike LaFleur had left McVay’s staff to become Arizona’s head coach.²¹

The spread option: words built for signals

The last family came from college football, and its language is shaped by one fact: it is designed to be *signaled*, not spoken. The college spread offense, and its run-heavy cousin the spread option, grew up in the 1990s and 2000s with coaches such as Rich Rodriguez, Urban Meyer and Chip Kelly. It puts the quarterback in the shotgun, spreads receivers wide to thin the box, and makes the quarterback a runner, so that a defender left unblocked can be “read” rather than blocked (Option Football).

Because these offenses wanted to play fast, and because college teams had no helmet radios until 2024, calls had to travel by hand signals and the big picture boards described in *What a Play Really Is*. That pushes the language toward **one-word or one-picture calls**, with a lot of the decision-making moved *after* the snap: the quarterback’s read decides whether it’s a handoff, a keeper or a throw. That last step, packaging a run and a pass into one call and letting the quarterback choose, is the RPO. The call can be short because the play is two plays.

The NFL’s most direct import was **Chip Kelly’s Philadelphia Eagles** (head coach 2013–15), who brought a college tempo offense,

¹⁹Kliff Kingsbury’s career (Texas Tech quarterback under Leach; Texas Tech head coach 2013–18, coaching Patrick Mahomes; Cardinals head coach 2019–22; Washington offensive coordinator 2024–25): https://en.wikipedia.org/wiki/Kliff_Kingsbury . No-huddle rates: course calculation in *What a Play Really Is*.

²⁰Tim Layden, *Blood, Sweat and Chalk* (2010), on Mike Shanahan’s years as 49ers offensive coordinator (1992–94) and the Broncos’ zone-running offense under him; Los Angeles Rams, “Sean McVay” coaching biography (Washington assistant 2010–16, including tight ends coach under offensive coordinator Kyle Shanahan, 2011–13; Rams head coach from 2017): <https://www.therams.com/team/coaches/> .

²¹2025 and 2026 staffs as compiled in this course’s fact sheet (checked 2026-10-06), §0 (C2) and §9, from team announcements and Pro Football Rumors, “2026 NFL offensive/defensive coordinator search tracker”: <https://www.profootballrumors.com/2026/02/2026-nfl-offensive-defensive-coordinator-search-tracker> . Kubiak was Seattle’s offensive coordinator for Super Bowl LX: https://en.wikipedia.org/wiki/2026_Seattle_Seahawks_season .

sideline signals and read-option runs to the league.²² The full system didn't last there, but its parts did: the shotgun read-option and the RPO are now in most NFL playbooks, and the Ravens' Lamar Jackson era and the Eagles' Jalen Hurts era built whole offenses around a quarterback who runs.

Why the language matters

Here is the whole comparison in one place.

Family	A call names...	Typical call (illustrative)	Built for	Gives up
West Coast	each player's job	"Gun Trips Rt 60, Z Go, H Out, Y Flat, X Dig"	precision, timing, fine detail	time, memory; awkward at tempo
Air Coryell	each route, by number	"Trips Rt 60, 619, H Out"	speed, vertical combinations	concepts live outside the call; tied to positions
Erhardt-Perkins	the whole concept	"Trips Rt 60, Ghost, X Dig"	portability, short calls, tempo	detail moves into rules; every player learns every role
Run-and-shoot	a route family; receivers choose	"Go," with motion	answers to every coverage, tiny playbook	protection; receiver-QB agreement
Air Raid	one of a few concepts	"Mesh" (often signaled)	repetition, tempo, simple teaching	variety; fewer under-center and heavy looks
Shanahan/McVay	a run look plus the play off it	about twenty words	identical starts for runs and passes	very long calls; huddle
Spread option	a package; the QB decides	one word or one sign	tempo, signals, QB run threat	quarterback hits; needs a running QB

The language shows up in three places a fan can actually see.

Installation: what can be taught by Sunday. Every offense has a fixed number of meeting and practice hours per week. A language that lets one word mean many plays (Erhardt-Perkins concepts, Coryell digits, the Air Raid's few plays) lets a team teach more *combinations* in that time; a language of individually named jobs (the WCO) lets it teach more *detail*. This is also why a new offensive coordinator matters so much more to players than fans expect: he often brings his language with him, and every player has to relearn the names of things he already knows how to do. A quarterback on his third coordinator in three years has learned three languages for mostly the same plays.

Tempo: how fast the call can travel. A call that has to be spoken in the huddle costs ten seconds. A call that fits in one word or one signal can be delivered at the line, which enables the no-huddle and, at the extreme, the hurry-up. It is no accident that the league's tempo offenses have spoken short languages: the 2012 Patriots with Erhardt-Perkins concepts, Kelly's Eagles with college signals, Kingsbury's teams with Air Raid habits. The tempo game itself is covered in Part 8.

In-game adjustment: how cheaply the plan can change. In the third quarter the offensive coordinator learns that a cornerback is jumping every out route. What can he change without a practice? In a WCO language, a one-word tag on the call ("Z Out-and-Up") changes just that route. In a digit system he changes one number. In a concept system he can keep the concept and move it to a new formation so the corner sees it from a different spot. And in every system the cheapest adjustments are **tags and formations**, not new plays, which is why the install is built the way it is.

How a system is installed

Install is the teaching of plays, and for a whole system it happens in layers. The order matters, because each layer is built from the one below it.

²²Pro Football Reference, "Chip Kelly" (Eagles head coach 2013–15): <https://www.pro-football-reference.com/coaches/KellCh0.htm> ; on college helmet radios arriving only in 2024, see the sources in What a Play Really Is.

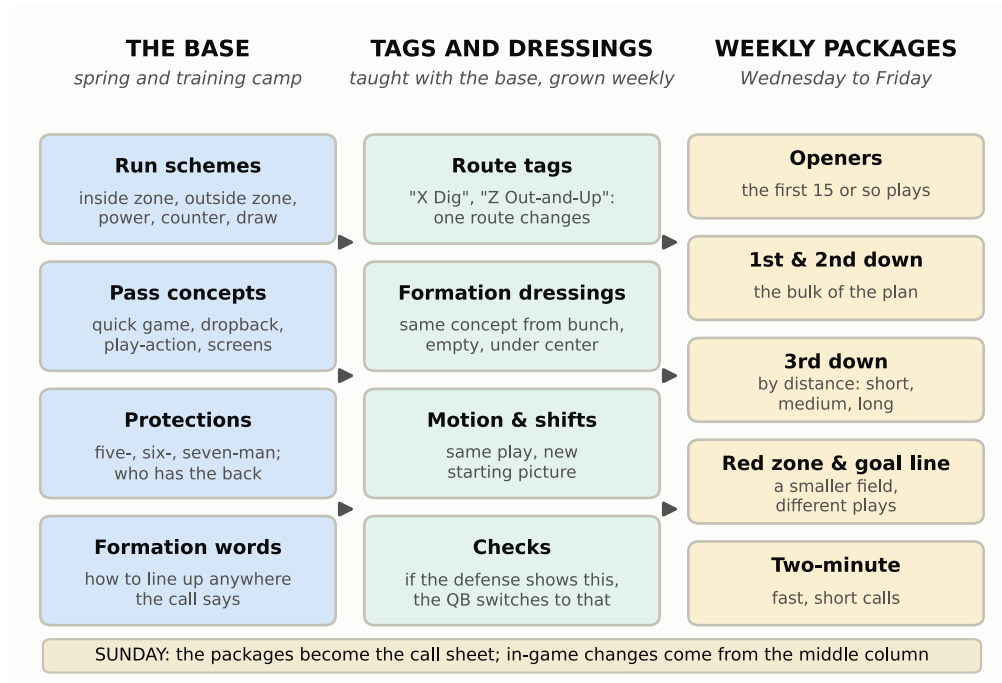


Figure 6: How a system is installed (a typical pattern; schedules vary by team). The base is taught in the spring and re-taught in training camp: the core runs, pass concepts, protections and the formation and motion vocabulary. Tags multiply the base into many variations without new plays. Each week the staff picks a slice of base-plus-tags for this opponent and installs it in packages by situation, which become the call sheet. On Sunday, adjustments are made with tags and formations, almost never with new plays.

Layer 1: the base. In the spring and again in training camp, the staff installs the system’s core, typically in numbered sessions (“Install 1” on the first day, “Install 2” on the second) that each add a few runs, a few pass concepts, a protection and some formations, and keep repeating what came before. The base is the system’s vocabulary: if a player doesn’t know it by September, he can’t play.

Layer 2: tags and dressings. Tags (from Part 1) and formation changes multiply the base without adding new plays. A concept taught as one play becomes a dozen once you can run it from four formations with three motions and two route tags. This layer is where offenses create the *appearance* of a giant playbook, and where the different languages differ most: Erhardt-Perkins was designed to make formation dressings cheap; the WCO makes route tags precise.

Layer 3: the weekly packages. Each week the staff studies the next opponent and chooses a slice of base and tags that should work against it, the game plan. It is installed by situation, on a schedule many NFL teams share: early downs first (Wednesday is often the “first-and-second-down day”), third downs next, then the red zone, goal line and two-minute plays at the end of the week. Each package becomes a box on the call sheet.

Two consequences are visible from the couch. First, **a new play in Week 10 is usually an old concept in new clothes**, a base pass from a formation the team hasn’t shown, not something invented that week. Second, **a team’s “new wrinkle” for the playoffs is usually a tag** (a receiver who used to run the out now runs the out-and-up), because a tag can be learned in a day without disturbing anything else.

! Why it exists: base, then tags, then packages

Why not just install the best plays for each opponent from scratch every week? Because players execute *learned* plays fast and new plays slowly, and in a sport decided by tenths of a second, “slow” means “wrong.” The layered install lets a staff put something new in front of the defense every Sunday while the players run things they have done hundreds of times. The cost is that the base must be chosen carefully in the spring: a concept that isn’t in the base can’t easily appear in November.

Fingerprints on film

You can't hear the call from the couch. But a system leaves fingerprints, because its language and beliefs push it toward certain looks over and over. Four are easy to read on any broadcast.

1. **Where the quarterback stands.** Under center points toward the Shanahan/McVay family and play-action; nearly always in the shotgun points toward spread, Air Raid or a pass-first WCO descendant.
2. **Motion.** Lots of motion, especially motion still moving at the snap, is a Shanahan/McVay and modern-WCO habit. Very little motion with lots of tempo suggests a spread or Air Raid lineage.
3. **Splits and empty.** Receivers lined up close to the formation (condensed splits) say the offense wants room to run crossing routes and to block on the edge; wide splits and frequent empty sets say "spread the field and throw."
4. **The huddle.** A team that routinely skips the huddle needs a short language. A team that huddles every snap and snaps late is probably speaking a long one.

Here is what two of those fingerprints looked like for every team in 2025.

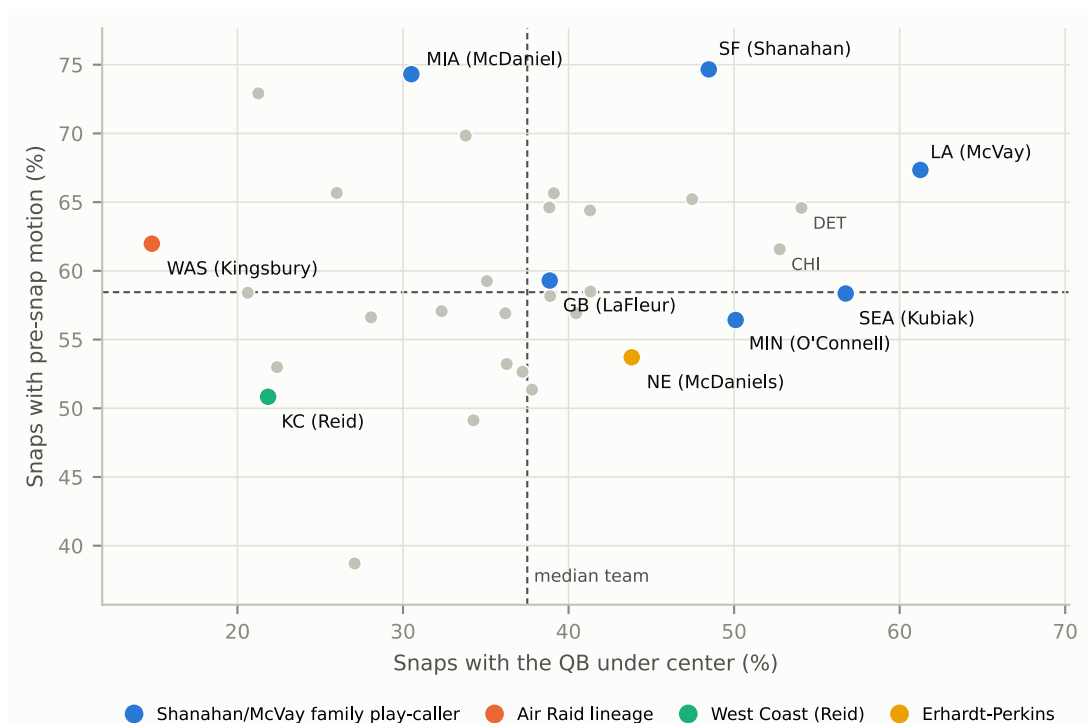


Figure 7: Two system fingerprints, 2025 regular season: the share of snaps with the quarterback under center (across) and the share with pre-snap motion (up), neutral situations only (outside the last two minutes of each half, win probability 20–80%). Blue: teams whose 2025 play-caller came from the Shanahan/McVay family; they cluster to the right (under center) and mostly high (motion). Orange: Washington under Kliff Kingsbury, the Air Raid heir, almost always in the shotgun. Also marked: Kansas City (Andy Reid, a West Coast descendant) and New England (Josh McDaniels, Erhardt-Perkins). Grey: the rest. Source: nflverse play-by-play joined to FTN charting (qb_location, is_motion); FTN counts motion before or at the snap.

The pattern is clear without being a law.²³ The Rams (McVay) were under center on about three snaps in five, the most in the league, with Seattle (Kubiak) second; the 49ers and Vikings were well above the median too, and the 49ers and Rams were among the league's heaviest users of motion. Washington, at the other extreme, was in the shotgun on all but about one snap in seven, and (as Part 1 showed) skipped the huddle on most of them. Andy Reid's Chiefs, a West Coast descendant crossbred with college spread ideas, lived in the shotgun too, which shows how far a family can drift from its roots. And the exceptions are instructive: the Bears and Lions were near the top in under-center snaps under Ben Johnson's influence, though he isn't a Shanahan-tree coach, and McDaniel's Dolphins used a Shanahan-family motion rate from the shotgun far more than their cousins did. Families are tendencies, not uniforms.

²³Course calculation from nflverse play-by-play joined to FTN charting (load_ftn_charting), 2025 regular season, runs and passes with a down, outside the last two minutes of each half and with win probability between 20% and 80%. Under center = qb_location == "U"; motion = is_motion, which FTN defines as motion "before or at the time of the snap" (so it is lower than PFF's shift/motion series and not comparable to it). Team codes follow nflverse (LA = Rams). Play-callers: course fact sheet, §9.

🔥 Common misconception

“They copied the 49ers’ playbook.” Teams copy *plays* constantly; every staff watches every other staff’s film, and a good new concept is in thirty playbooks by the next season. What is hard to copy is the system underneath: the language that makes the play cheap to call, the teaching that makes it automatic, and the dozens of related plays that make it hard to defend. That’s why coaches are hired for their system and why their assistants are hired away: the system travels with people, not with diagrams.

💡 Watch for it on Sunday: fingerprint the system

Pick one offense and chart one drive per quarter (eight to fifteen snaps is enough to start). For each snap, write:

- **U / G / P**: quarterback under center, in the shotgun, or in the pistol (in between).
- **M**: did anyone move before the snap? Mark **M!** if he was still moving at the snap.
- **C / W / E**: receivers condensed (tight to the formation), wide, or an empty backfield.
- **H / N**: huddle or no-huddle.
- **Run, pass or play-action**, from the first step of the line and the quarterback’s fake.

Then guess the family. Mostly U, lots of M!, condensed splits and play-action: the Shanahan/McVay dialect. Mostly G with wide splits, empty sets and little huddling: Air Raid or spread lineage. Shotgun, quick timing throws, lots of formations but the same few two- and three-man combinations recurring: an Erhardt-Perkins or modern WCO offense. Log your guess with the team’s play-caller and season in your prediction log, then check it against the chapter’s chart or the team’s coaching history. The fun part is the mismatch: a team that doesn’t look like its tree is usually telling you something about its quarterback.

Predict the play

💡 Predict the play 1: the slot’s choice

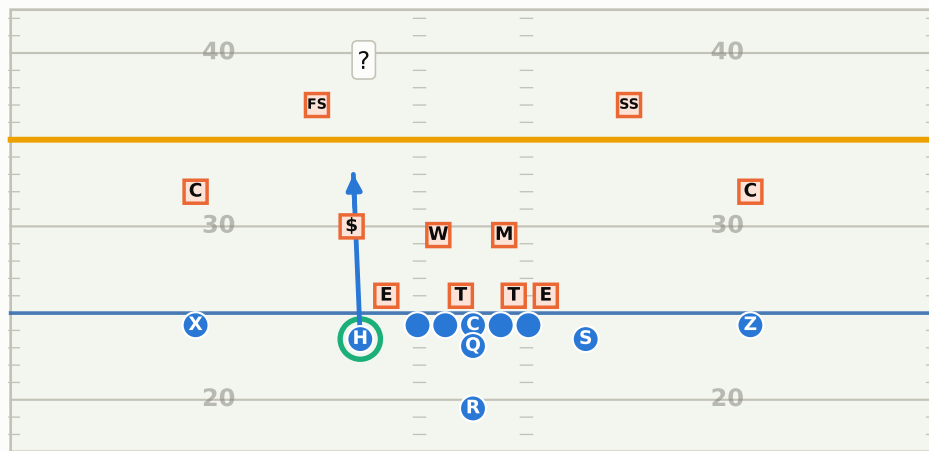


Figure 8: Run-and-shoot base against a nickel defense. Before the snap the defense shows two deep safeties, each about 12 yards deep and outside the hashes. The left slot (H, ringed) runs a choice route: a vertical stem with a decision at about 10 yards.

Which branch should H take, and where is the quarterback likely to throw?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: whose language is this?

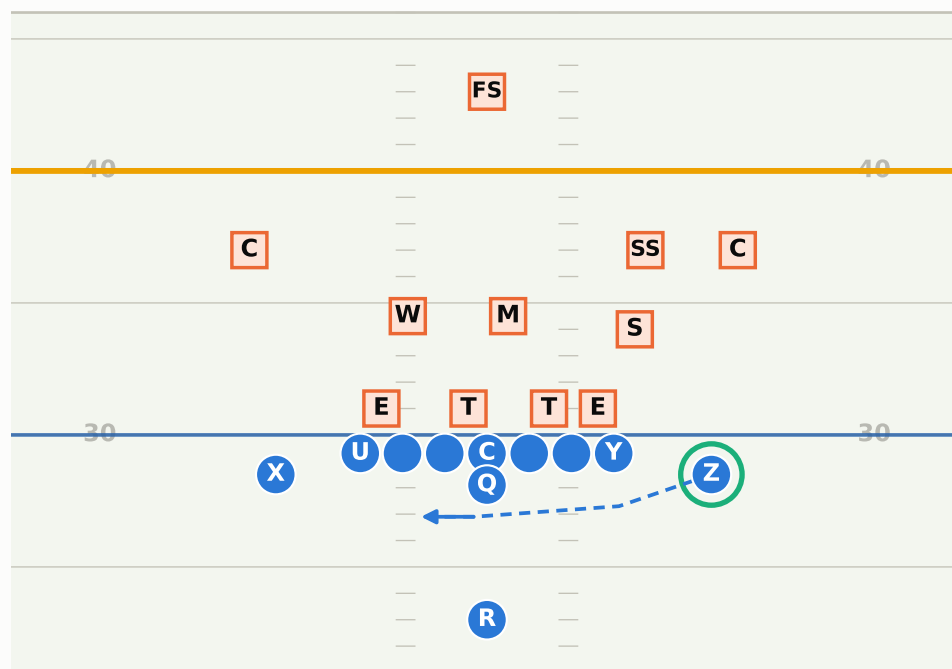


Figure 9: First-and-10 on the offense's own 30. Two tight ends (12 personnel), quarterback under center, both wide receivers in tight splits about eight yards from the ball. Before the snap, Z comes in jet motion across the formation, still moving as the ball is snapped (dashed).

Which offensive family does this look come from, and what are the two or three most likely plays?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: one word, new formation

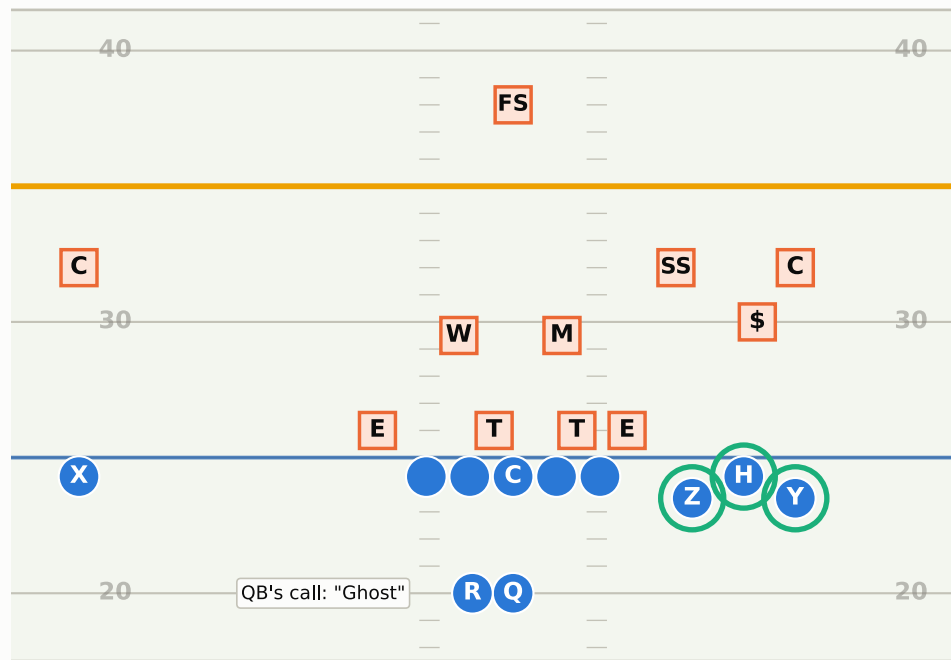


Figure 10: Earlier in the drive the offense ran ‘Ghost’ from trips right, with the tight end (Y) running to the flat. Now it lines up in a bunch to the right: the tight end is the widest man in the bunch, H is on the line in the middle of it, and Z is the innermost. The quarterback’s call at the line is the same word: ‘Ghost’.

Who runs the vertical, who runs the eight-yard out, and who runs to the flat?

Answer 3 is at the end of the chapter. Try it first.

Checklist so far (Parts 1–4)

Part 4 ends here, so it’s time to update the pre-snap card from Part 2. The new lines are about the offense: where the quarterback stands, how many players stay in to block, what the line’s first step says at the snap, and, now that you know the families, whose language the offense speaks. The card grows again after the defensive parts and becomes the master checklist of the capstone.

Watching checklist · after Part 4

new in Parts 3–4

BEFORE THE SNAP

- 1 Situation: down, distance, field zone, clock and score Part 1
- 2 Personnel: who is on the field for each side 02-01, 02-05
- 3 Formation and strength: 2x2, 3x1, bunch, empty; where is the tight end? 02-02 to 02-04
- 4 Motion: does a defender travel with it (man) or do they bump over (zone)? 02-06
- 5 Box count: blockers against defenders near the line 02-05, 03-01
- 6 Shell, first look: one deep safety or two? 02-05
- 7 QB depth: under center (run and play-action lean), pistol, or shotgun (pass lean) Parts 3–4
- 8 Protection count: does the back or tight end look set to stay in and block? 04-02
- 9 System fingerprint: under center %, motion at the snap, splits, huddle; whose family? 04-08

AT THE SNAP

- 10 Line's first step: run block, pass set, or a run fake (play-action)? 03-01, 04-05
- 11 RPO? At the mesh, find the defender the quarterback is looking at 04-06

TWO SECONDS AFTER

- 12 Concept family: horizontal stretch, vertical stretch, triangle, quick game (ball out by ~2.5 s)? 04-03, 04-04
- 13 QB's eyes: how many reads before the throw, or did he leave the pocket? 04-07
- 14 Log it: your pre-snap guess against what happened 01-06

Figure 11: The watching checklist after four parts of the book. Lines 1–6 come from Parts 1 and 2; the highlighted lines are new in Parts 3 and 4: the quarterback's depth, the protection count, the system fingerprint, the line's first step, the RPO tell, the concept family and the quarterback's eyes.

Takeaways

- **A system is a belief, a core of plays, a language and a teaching method.** It isn't a formation or a list of plays; two teams can run the same play inside very different systems.
- **The three classic NFL languages make different bets.** The West Coast names each player's job (precise, long); Air Coryell numbers each route (fast, position-bound); Erhardt-Perkins names whole concepts that work from any formation (portable, short, rule-heavy). Most modern playbooks mix them.
- **Other families moved decisions after the snap.** The run-and-shoot lets receivers choose routes by coverage; the Air Raid wins with a tiny playbook and endless repetition; the spread option and the RPO let the quarterback choose between a run and a pass. Each needs less in the call because more happens in the play.
- **The language shows up on TV** as tempo (short calls allow the no-huddle), as formation variety (concept words make new looks cheap), and as how a team adjusts mid-game (tags and formations, rarely new plays).
- **Installation is layered: base, then tags, then weekly packages.** That's why a "new" play is usually an old concept in a new formation, and a playoff wrinkle is usually a tag.
- **Fingerprints, not labels.** Under-center rate, motion at the snap, splits, empty and huddle habits point to a family, but families drift; label every staff with its season.

Connections

Builds on: What a Play Really Is (the anatomy of a play call, verbiage, tags and install), Passing Fundamentals (the route tree and option routes), Dropback Concepts (the concepts these languages name, including the Air Raid's mesh and Y-cross), Run-Pass Options and Quarterback Play.

Sets up: the defense's side of the same story in Parts 5–7, and then Concepts Versus Coverages, The Game Plan (the weekly install from the staff's side) and In-Game Adjustments, Tempo, and the Sideline Battle. Where the systems came from is history: 1978 and the Passing Revolution (Walsh, Coryell, the rule changes that let them flourish) and The College Laboratory (the run-and-shoot, the Air Raid and the spread option).

Case studies: Walsh's 49ers and the West Coast Offense (the WCO in full, and the coaching-tree map) and The Shanahan Wide-Zone Tree; also Belichick's Patriots for the Erhardt-Perkins game plan and Andy Reid's Chiefs for a West Coast descendant crossbred with college ideas. The trees themselves are mapped in Appendix E.

Go deeper

- Chris B. Brown, "Speak My Language," *Grantland* (2013), and *The Art of Smart Football* (2015): the clearest short treatment of play-calling languages, and of the run-and-shoot's legacy.
- Bill Walsh with Brian Billick and James Peterson, *Finding the Winning Edge* (1998): Walsh's own manual on installing and running an offense, written for coaches.
- S. C. Gwynne, *The Perfect Pass: American Genius and the Reinvention of Football* (2016): Hal Mumme, Mike Leach and the birth of the Air Raid.
- Tim Layden, *Blood, Sweat and Chalk* (2010): chapters on the West Coast offense, the run-and-shoot and the spread, told through their inventors.

Answers to Predict the play

Answer 1.

Split the safeties: H keeps running, bending slightly toward the middle, between the two deep safeties, and the quarterback throws it there. With two deep safeties each responsible for a half of the field, the *middle* of the deep field is the seam between their halves. Neither safety can play H there without abandoning the receiver outside him, so if the slot gets past the underneath defenders the window opens about 15–20 yards downfield.

Against one deep safety (the animation earlier in the chapter) the same route would be wrong: running into the middle takes H straight to that safety. So he breaks it off in front of him. The quarterback must reach the same conclusion from the same safeties; that agreement is the whole run-and-shoot. If the defense *rotates* after the snap (two-high before, one-high after), both of them have to see the rotation in time, which is why defenses facing the run-and-shoot leaned on disguise (Disguise, Rotation, and Split-Field Coverage).

Answer 2.

The Shanahan/McVay family, and the likely plays are an outside-zone run, a play-action pass off it, or a bootleg the other way. Every fingerprint points the same way: under center, two tight ends, condensed splits and jet motion at the snap. The condensed receivers are close enough to block on the edge of a run *and* to run crossing routes behind linebackers who step up for the fake. The jet motion pulls defenders sideways and can itself become a sweep or a decoy.

The point of the look is that you *shouldn't* be able to tell which of the three it is until a second after the snap. Watch the linebackers' first step, then the quarterback: if he turns his back to the defense, fakes the handoff and keeps the ball, look for the crosser coming open behind those linebackers (Play-Action, Bootlegs, and Screens). If you guessed "run," you're right more often than not from this look on first down, but the whole system is built so that guess costs the defense.

Answer 3.

The tight end runs the vertical, H runs the out, and Z runs to the flat. Ghost assigns roles by position in the formation, counted from the outside in: #1 (here Y, the widest) goes vertical, #2 (H) breaks out at about eight yards, #3 (Z, the innermost) releases straight to the flat. On the previous Ghost, the tight end was #3 and ran the flat; now he is #1 and runs the deep route.

That is concept-based terminology doing its job. The defense sees a formation it may not have prepared for, a tight end running a vertical and a receiver running the flat, while the offense runs a play every player has known since the spring. The bunch adds a

bonus: the receivers start so close together that defenders playing man coverage have to fight through one another to follow them (Spread Formations).